

PRO192- Practice Exam – SU24

Duration: 85 minutes

Develop Product management OOP application with the following components and requirements

1. Create an MVC console application in NetBeans:

a. Define class Product with the following attributes and methods: (2 marks)

- + ProductID, ProductName, Category, Price, ProductDate.
- + Constructors, get/set(), toString()
- + Method for getdiscount(1 mark):
 - If the age of ProductDate is less than 1: no discount
 - If the age of ProductDate is less than 2: discount 20% price
 - If the age of ProductDate is bigger 2: discount 50% price

Note: No IO console is used in this model class

b. Define class ProductList with the following attributes and methods: (3.5 marks)

- + List of Product
- + Read text file to load data for ProductList– product.txt is provided
- + Method that can search product by any criteria
- + Method sort products in descending order by price
- + Method Statistics on products of the same Category

Note: No IO console is used in this model class

c. Develop ProductManagement class for application(3.5 marks)

- ❖ The main menu for the application includes: Displaying all products, display all promoted products, Searching products, Sorting lists of products, Statistic products of the same category.

Hint: An abstract Menu class is provided for use in the application. Override method execute() call the methods below:

- + Methods display all products can apply any product list
- + Methods display all promoted products
- + Method for searching product with searching menu allow user select search criteria
- + Call sort products in descending order by price
- + Call statistic products of the same category method

- ❖ main() method to run the program.

Note: This class allows interaction with user IO from the console.

2. Technical requirements(1 mark):

+ Create package model for Product and ProductList class, create package view for Menu, create package controller for ProductManagement class

+ Checking for ProductID has 4 characters and starts with "P" (Example: P001).

Error message: "The ProductID is in wrong format (start P and has 4 chars)".

+ User interface as sample images below:

```
Product Management System
-----
1.List all Product
2.Promotional products
3.Product Search
4.Sort Product
5.Product statistics
6.Exit
-----
Enter selection..1
List of Product
-----
P002 | Shampoo | Haircare | 12.5 | 10/03/2023
P003 | Lipstick | Makeup | 10.59 | 15/02/2021
P004 | Body Lotion | Bodycare | 18.75 | 20/04/2023
P005 | Sunscreen | Skincare | 22.0 | 30/05/2022
P006 | Conditioner | Haircare | 13.99 | 25/06/2024
-----
Total :5
```

Enter selection..2

List of promotional products

P002	Shampoo	Haircare	12.5	10/03/2023	0.20
P003	Lipstick	Makeup	10.59	15/02/2021	0.50
P004	Body Lotion	Bodycare	18.75	20/04/2023	0.20
P005	Sunscreen	Skincare	22.0	30/05/2022	0.20

Enter selection..3

ProductSearching

1.Find by Category

2.productionDate

3.List products with price>15

Enter selection..1

Enter Category: : Haircare

Searching Result

P002	Shampoo	Haircare	12.5	10/03/2023
P006	Conditioner	Haircare	13.99	25/06/2024

Total :2

Enter selection..2

Enter production Date : : 30/05/2022

Searching Result

P005	Sunscreen	Skincare	22.0	30/05/2022
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Total :1

Enter selection..3

Searching Result

P004	Body Lotion	Bodycare	18.75	20/04/2023
P005	Sunscreen	Skincare	22.0	30/05/2022

Total :2

Product Management System

- 1.List all Product
- 2.Promotional products
- 3.Product Search
- 4.Sort Product
- 5.Product statistics
- 6.Exit

Enter selection..4

Sorted product

P005	Sunscreen	Skincare	22.0	30/05/2022
P004	Body Lotion	Bodycare	18.75	20/04/2023
P006	Conditioner	Haircare	13.99	25/06/2024
P002	Shampoo	Haircare	12.5	10/03/2023
P003	Lipstick	Makeup	10.59	15/02/2021

Total :5

Enter selection..5

The products are same category

P006	Conditioner	Haircare	13.99	25/06/2024
P002	Shampoo	Haircare	12.5	10/03/2023

Total :2

Notice: Student must strictly follow all requirements as described above for marking, Any modification of the requirement detail is not allowed and will get no mark!